

Q. 1 Let x, y, z be positive integers such that $\gcd(x, y, z) = 1$, and the expression

$$\frac{xy}{z} + \frac{yz}{x} + \frac{zx}{y}$$

is an integer. Prove that xyz is a perfect square.

Q. 2 Prove that for any positive integer $n > 1$, the equation

$$1 + 2x + 3x^2 + \cdots + nx^{n-1} = n^2$$

has a rational root between 1 and 2.

Q. 3 Let $f : [0, 1] \rightarrow \mathbb{R}$ be a continuous function such that

$$\int_0^1 f(x) dx = \int_0^1 xf(x) dx = 1.$$

Prove that

$$\int_0^1 (f(x))^2 dx \geq 4.$$

Q. 4 Consider three distinct points A, B, C on the rectangular hyperbola $xy = 1$. Prove that the orthocentre of $\triangle ABC$ also lies on the hyperbola.

Q. 5 Let n be any natural number. Define

$$S_n := \frac{1}{n+1} \sum_{k=0}^n \frac{1}{\binom{n}{k}}.$$

Answer the following:

- (a) Prove that, $2S_{n+1} - S_n = \frac{2}{n+2}$, for all $n \in \mathbb{N}$.
- (b) Using part (a) or otherwise, prove that, for all $n \in \mathbb{N}$,

$$S_n = \frac{1}{2^{n+1}} \sum_{k=1}^{n+1} \frac{2^k}{k}.$$

Q. 6 Let $f : (0, \infty) \rightarrow \mathbb{R}$ be a differentiable function such that $\lim_{x \rightarrow \infty} f'(x) = 0$. Prove that $\lim_{x \rightarrow \infty} (f(x+1) - f(x)) = 0$.

Q. 7 On the quadratic polynomial $P(x) = ax^2 + bx + c$, you are allowed to make two types of transformations:

- (a) You can swap a and c .
- (b) You can replace x with $x + t$, where t can be any real number.

Prove that, applying these transformations repeatedly (in any order, any number of times), you can never convert the polynomial $x^2 + x + 1$ to the polynomial $x^2 + 1$.

Q. 8 Suppose you are given two consecutive perfect cubes whose difference is a perfect square, say k^2 . Then show that, k must be the sum of two consecutive perfect squares. (For example, observe that: $8^3 - 7^3 = 169 = 13^2$. Furthermore, $13 = 2^2 + 3^2$.)

-End of the Question Paper-